UK HEAT SPLIT

Data sources and estimation methodology

Causeway Energies · causewaygt.github.io/uk-heatsplit · v7.1.0 · August 2026 · contact@causewaygt.com

Abstract

UK Heat Split is a public, automatically updated estimate of the weekly heating and cooling energy balance of Great Britain. Daily gas throughput to the low-pressure distribution network is decomposed into a temperature-sensitive space-heating component and a weather-independent baseline by segmented linear regression on population-weighted degree days, following the established gas-to-heat method of Watson et al. [1,2] and Sansom [3]. The trailing twelve-month space-heating estimate is calibrated against the Department for Energy Security and Net Zero (DESNZ) Energy Consumption in the UK (ECUK) end-use tables [4]. Non-gas fuels and space cooling, which lack daily operational feeds, are held at annual ECUK levels and modulated by degree days. Delivered service is derived from purchased energy using in-situ combustion efficiencies [5] and measured seasonal heat-pump performance factors [6,7]. Cooling demand in the unequipped building stock is estimated independently from overheating degree-hours above the CIBSE comfort threshold [8] and dwelling-stock statistics [9,10]. All emissions use DESNZ conversion factors [11]; all monetary values use the prevailing Ofgem price cap [12]. The headline estimates are additionally published as a weekly historical series over the window the live gas feed serves, priced and carboned at the rates in force each week. Every figure resting on author judgement is flagged; the complete source code and data are public. This document specifies each estimator, its inputs, its assumptions, and its uncertainty.

1. Conventions

1.1 Sourced quantities and author estimates

Two classes of number appear on the dashboard. Sourced quantities are traceable to a cited publication or a live public feed. Author estimates (marked † on the site) rest on Causeway Energies judgement where no authoritative figure exists; each is accompanied by a stated range and an invitation to challenge. The distinction is maintained throughout this document.

1.2 Purchased energy and delivered service

Two physical quantities are tracked and must not be conflated. Purchased energy is metered fuel or electricity at the point of supply. Delivered service is the useful heat or cooling supplied to the conditioned space after conversion. For a combustion appliance of efficiency $\eta < 1$, delivered service is less than purchased energy; for a heat pump of coefficient of performance $COP > 1$, delivered service exceeds purchased electricity because ambient or ground heat, never purchased, is harvested. The indigenous-supply and efficiency arguments on the site depend on this distinction.

2. Primary heating estimate: gas-to-heat regression

2.1 Data source

The primary feed is the National Gas Transmission operational-data REST API [13], from which the daily aggregate demand of the thirteen Local Distribution Zones (LDZ) is retrieved and summed. Publication identifiers are resolved from the API catalogue at runtime rather than hard-coded, so that a renamed publication produces a diagnostic rather than a silent error. The total National Transmission System (NTS) demand is retrieved separately for context. The LDZ aggregate is used in preference to the NTS total because the latter includes gas-fired power generation and directly connected industry, which are unrelated to building heat and would dominate the signal.

2.2 Quantifying the LDZ / NTS distinction

The distinction is material. In the current week the summed LDZ demand is of order 480 GWh/day at summer minimum, against an NTS total of order 1,000–1,400 GWh/day; the difference is principally power-station and industrial offtake carried on the transmission system, which does not track heating degree days. Using the NTS total would introduce a large, weather-insensitive term and depress the apparent temperature sensitivity of demand.

2.3 Temperature index

Outdoor temperature is represented by a population-weighted mean over twelve Great British urban centres (London, Birmingham, Manchester, Leeds, Glasgow, Newcastle, Bristol, Cardiff, Edinburgh, Southampton, Nottingham, Sheffield), with weights from Office for National Statistics mid-year population estimates. Hourly and daily 2 m air temperatures are drawn from the ERA5 reanalysis [14] via the Open-Meteo archive API [15]. Heating degree days at base temperature T_e are computed as HDD = max(0, $T_e - \bar{T}$), with $\bar{T}$ the weighted daily mean; cooling degree days use CDD = max(0, $\bar{T}$ − T_c). Bases are evaluated at 14.5, 15.5 and 16.5 °C for heating and 18 °C for cooling; the heating base is selected on goodness of fit.

2.4 Estimator

Daily LDZ demand G is modelled as a segmented (piecewise-linear) function of temperature, equivalent to an ordinary-least-squares regression on the degree-day transform:

$$G_i = \alpha + \beta \cdot HDD_i + \varepsilon_i$$

where α (GWh/day) is the weather-independent baseline attributed to water heating, cooking and other non-space-heating gas use; β (GWh per degree day) is the marginal space-heating response; and ε is a residual. The fitted space-heating component on day i is $\beta \cdot HDD_i$, and the baseline is α. This is the decomposition of Watson et al. [1,2], itself building on Sansom [3]; the method is standard in the GB heat-demand literature. Current fitted parameters are α ≈ 476 GWh/day, β ≈ 130 GWh/HDD, with base T_e = 16.5 °C over a 365-day window (Figure 1).

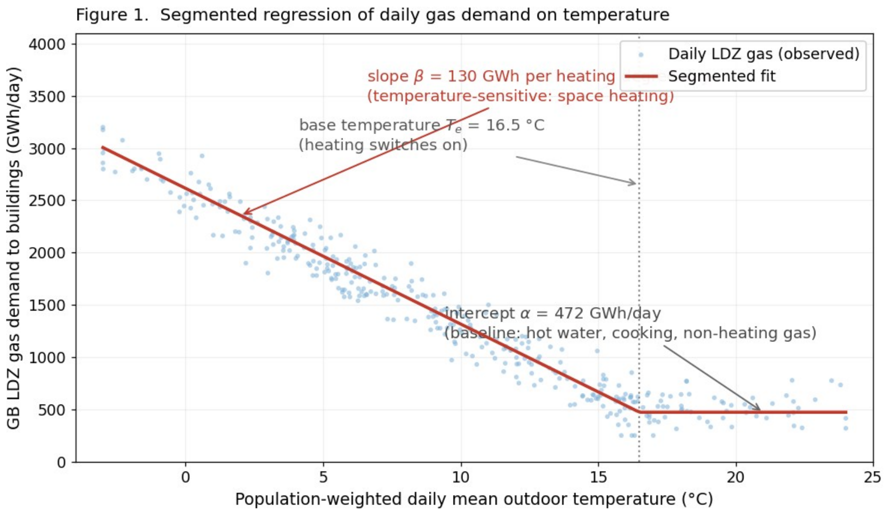

Figure 1. Segmented regression of daily gas demand on temperature. Schematic illustration on live parameters.

2.5 On the coefficient of determination

The regression returns $R^2 \approx 0.92$. Two caveats attach. First, because HDD and demand share a strong annual cycle, a high R^2 is expected and does not by itself validate the functional form. Second, R^2 is inflated by the seasonal range; the residual standard error (of order 180 GWh/day) is the more informative figure for daily uncertainty and is reported in the calibration diagnostics.

3. Calibration to national statistics

The regression yields a self-consistent daily signal but not an absolute national total. The trailing twelve-month integral of the fitted space-heating component is therefore reconciled against ECUK [4], which reports domestic and non-domestic energy consumption by end use and fuel, derived by DESNZ from the Digest of UK Energy Statistics (DUKES) national energy balance combined with the Cambridge Housing Model and the National Energy Efficiency Data-Framework. ECUK space-heating totals are themselves modelled estimates, weather-corrected to a long-run average using the same degree-day apparatus, not metered heat.

The calibration ratio r = (modelled 12-month space heat) / (ECUK anchor, GB-adjusted and weather-normalised) is monitored continuously. The publication gate is $|r - 1| \leq 0.10$. The current value is $r \approx 1.10$, i.e. the live model runs about 10% above the ECUK anchor — at the edge of tolerance, and reported openly on the site rather than silently rescaled. Candidate contributions to the residual include the treatment of non-domestic gas, the fixed base temperature, and weather-normalisation differences between the trailing window and the ECUK reference year.

Figure 2. Within-class centring (month × weekend) isolates the cooling anomaly

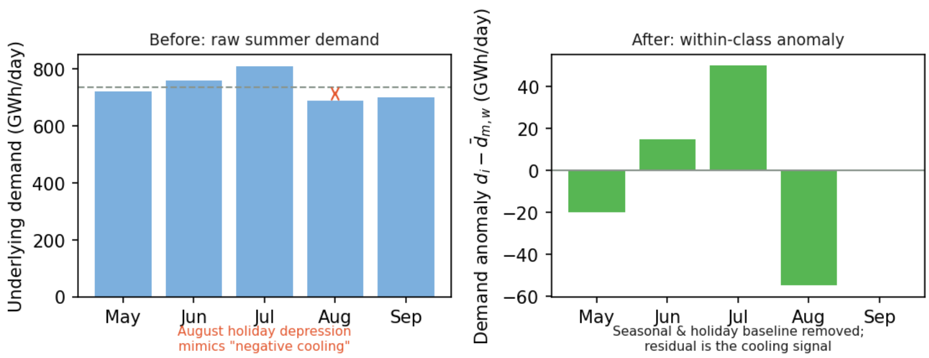

Figure 2. Within-class centring. Raw summer demand (left) is depressed in August by the holiday reduction in commercial and industrial activity, which a naive regression misreads as negative cooling. Subtracting the month × weekend-class mean (right) removes the seasonal and holiday baseline, leaving the cooling anomaly.

4. Cooling: the equipped fleet (observed)

4.1 Data source and reconstruction

Cooling delivered by installed plant is inferred from national electricity demand. Half-hourly demand is taken from the National Energy System Operator (NESO) historic demand data and rolling update files [16]. Metered demand understates true underlying demand by the quantity supplied behind the meter, principally embedded solar photovoltaics, which is largest precisely on the hot, bright days when cooling load peaks. Underlying demand is therefore reconstructed by adding NESO estimates of embedded solar and wind generation to metered demand before analysis. The correction is material: embedded solar alone reaches of order 10–16 GW at midday in high summer against a metered demand of order 20–30 GW, so the hot, bright hours in which cooling peaks are precisely those the meter understates most. Future-dated rows present in the update file, and days with other than 48 half-hourly values, are removed.

4.2 Within-class centring

Summer electricity demand carries a strong non-cooling structure: it falls during the August holiday period and differs between weekdays and weekends. Binning raw demand against CDD conflates this structure with the cooling response and can yield a spurious negative uplift. To isolate the cooling signal, each daily underlying-demand value d_i is expressed as an anomaly about the mean of its (calendar-month × weekend/weekday) class:

$$a_i = d_i - \bar{d}(m_i, w_i)$$

where $\bar{d}(m, w)$ is the mean underlying demand over all days sharing month m and weekend/weekday status w. This within-class centring (Figure 2) removes the seasonal and holiday baseline while preserving the temperature-driven excursion. The operation is the within transformation of standard panel econometrics — a two-way fixed-effects demeaning on calendar month and weekend/weekday class [31] — applied here to isolate a weather signal rather than to estimate a panel coefficient: the estimator is textbook, its application to GB cooling demand is this site's own, and it is validated against synthetic data with a known injected slope in

Section 4.4. The term “demeaning” is avoided in public-facing copy in favour of “within-class centring” to prevent confusion with the ordinary meaning of the word.

4.3 Response curve and latent slope

The anomalies a_i are grouped into CDD bins; the mean anomaly per bin is the observed cooling response (Figure 4). A linear slope fitted to the low-CDD bins, where the fleet operates unconstrained, gives the latent response in GWh per CDD; extrapolating it defines the demand that the equipped fleet would meet in the absence of capacity or behavioural limits. Departure of the observed curve below this line at high CDD would indicate saturation. The current curve does not saturate — the highest bin shows the largest uplift — consistent with field evidence that the installed UK cooling stock is not capacity-constrained at present temperatures, and implying headroom rather than a ceiling.

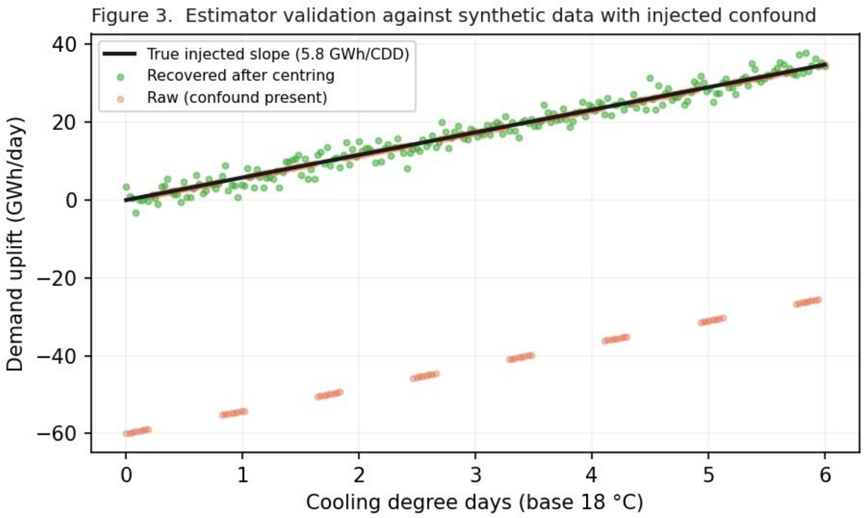

Figure 3. Estimator validation. A synthetic demand series was constructed with a known cooling slope and a holiday depression deliberately injected. The raw binning (orange) is biased downward by the confound; within-class centring recovers the true slope (green vs. black).

4.4 Validation

The centring estimator was validated before deployment on synthetic data in which a known cooling slope was combined with an injected holiday depression of comparable magnitude to that seen in real August demand (Figure 3). Naive binning of the synthetic series reproduced the spurious negative-uplift artefact; within-class centring recovered the injected slope to within sampling error. This confirms that the procedure removes the confound it is designed to remove without distorting the underlying signal.

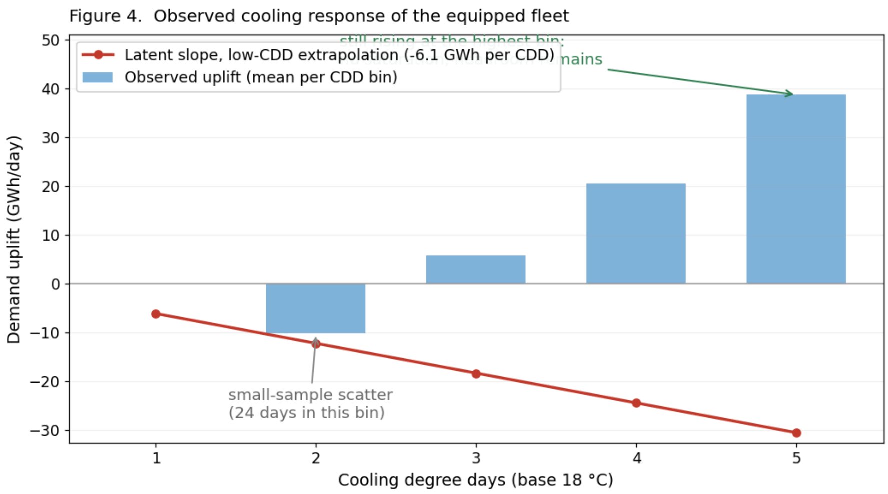

Figure 4. The live cooling response curve (this is the chart shown on the site). Bars are the mean demand uplift per CDD bin; the line is the latent low-CDD slope extrapolated. The negative third bin is small-sample scatter (ten days); the continued rise at the highest bin indicates no saturation.

4.5 Relationship to the billed cooling figure

The observed electricity response yields a weekly cooling electricity of order 150–230 GWh, whereas the ECUK-anchored figure used in the cost and carbon panels is of order 1,000–1,200 GWh. This order-of-magnitude discrepancy is stated on the site and left unreconciled pending a full cooling season. Both estimates may carry error. The observed figure captures only the marginal, weather-driven increment in national demand and will *understate* cooling to the extent that base-load cooling (data centres, process refrigeration, continuously conditioned commercial space) is weather-insensitive and absorbed into the class means. The ECUK-shaped figure, conversely, distributes an annual national cooling estimate across the year by CDD and may *overstate* a given summer week if the annual ownership or intensity assumptions are high, or if the CDD base temperature (18 °C) misrepresents the temperature at which conditioning actually begins. From v5.2 the planned resolution runs as a published diagnostic: a year-round ordinary-least-squares fit of daily underlying demand on HDD and CDD jointly — the CDD base swept from 10 to 18 °C and chosen on fit, so the data decides where conditioning starts — with weekend, August-holiday and Christmas-period dummies and a linear trend. The fitted cooling slope multiplied by trailing twelve-month CDD bounds annual weather-driven cooling electricity; the remainder against the ECUK anchor is weather-flat, an upper bound on ventilation plus base-load cooling plus below-base conditioning that this method cannot separate (†). The diagnostic has run continuously since v5.2 and is published today inside the cooling panel's evidence fold, alongside the summer centring slope, as an internal consistency check. It does not drive the bill or carbon panels. The switchover decision is scheduled for September 2026, once a full observed cooling season (May–September) is in hand; if taken, it would restate the summer cooling figures and the shares derived from them by the same documented restatement mechanism used for the weekly history.

5. Cooling: the comfort deficit (modelled)

The observed method sees only buildings that possess cooling. Since fewer than 5% of UK dwellings are air-conditioned [17], the electricity signal is structurally blind to the majority of the population experiencing summer overheating. This latent demand is therefore estimated independently, and explicitly as a stock-and-engineering calculation rather than an inference from the electricity data.

Overheating exposure is quantified as overheating degree-hours above the CIBSE operative-temperature threshold of 26 °C for occupied spaces [8], population-weighted, from the same hourly ERA5 series:

$$ODH_{26} = \Sigma_h \max(0, T_h - 26)$$

summed over hours h in the week. All hours are counted; an occupied-hours weighting (a candidate refinement, †) would reduce the total. The 26 °C threshold is a comfort-risk criterion, not a thermostat setting: newly installed air conditioning is typically set nearer 22–24 °C and would run for materially more hours than this tier counts. The direction is unambiguous — a lower switch-on temperature raises both hours and load — and the scale is large: relaxing the threshold from 26 to 24 °C roughly doubles the overheating degree-hours of a typical GB summer week (†), with load and cost rising in proportion. This tier is therefore a conservative floor on latent demand rather than a central estimate of what installed equipment would draw. The latent thermal demand is ODH_{26} scaled by the affected, unequipped dwelling stock and a per-dwelling thermal-response coefficient, plus an equivalent term for uncooled non-domestic comfort floor area derived from the Building Energy Efficiency Survey [18].

5.1 Uncertainty on the stock and response assumptions

The dominant uncertainties are the overheating-prevalence fraction and the thermal-response coefficients, all author estimates (†). Three prevalence scenarios are carried and never averaged: a low case of 11% of dwellings, from English Housing Survey self-reported overheating [9]; a high case of over one-half, from the Climate Change Committee assessment of dwellings at overheating risk [10]; and a central case of 25% (†). The per-dwelling response (order 0.2 kWh per degree-hour †) and the non-domestic response (order 3 Wh m^{-2} per degree-hour †, over ~190 Mm^2 †) are engineering judgements with a plausible factor-of-two spread each. These ranges bound the latent-demand estimate; they are independent of, and do not narrow, the observed-versus-ECUK discrepancy of Section 4.5, which concerns the equipped fleet rather than the deficit.

5.2 Delivery efficiency and the geocooling comparison

The central latent thermal load is expressed as the electricity required to meet it by two routes: passive or near-passive ground cooling at an effective COP of order 20 (†), and air-source vapour-compression at a seasonal EER of order 3.5. The resulting six-to-one ratio quantifies the electrical advantage of ground-coupled cooling. The seasonal-storage corollary — that heat rejected to an aquifer or borehole field in summer is recoverable in winter at a round-trip thermal efficiency of order 70% (literature range 50–80%) [19] — is shown in the dashboard's seasonal-mirror graphic with the loss represented explicitly.

6. Conversion, cost and carbon

6.1 In-situ efficiencies and performance factors

Combustion appliances are derated by in-situ, not nameplate, efficiencies: gas 0.835 and oil ~0.82 [5]. Heat pumps are credited with a seasonal performance factor of 3.24 for ground-source systems, from the Energy Systems Catapult in-situ monitoring of more than 1,700 installations [6]; the associated cold-weather stability of ground-source relative to air-source is documented in the Electrification of Heat demonstration [7]. These are the values underlying the delivered-service bars and the daily spark gap.

6.2 Cost

Retail costs are priced by sector. Each fuel component is split into its domestic and services shares using the same ECUK U3/U5 tables that anchor the mix; the domestic share is priced at the prevailing Ofgem default-tariff cap unit rates [12], refreshed quarterly, and the services share at non-domestic rates anchored on the DESNZ Quarterly Energy Prices non-domestic tables [30] (Causeway estimates †, set within the published size-band range — the Q4 2025 manufacturing averages of 3.5 p/kWh gas and 16.8 p/kWh electricity excluding CCL form the large-user floor, with services buildings sitting in the higher small and medium bands — refreshed annually). Cooling, an ECUK services-only category, is priced wholly at the non-domestic electricity rate. Standing charges are excluded throughout. The daily spark gap uses wholesale commodity prices only — National Gas System Average Price [13] and the Elexon market index [20] — divided by the respective efficiency or performance factor, and excludes network, policy and supply costs. Oil is priced from published heating-oil surveys (†).

6.3 Carbon

Combustion emissions use DESNZ (Defra) greenhouse-gas conversion factors [11], reported on a gross calorific value basis consistent with UK metered gas billing: natural gas 0.18296 $kgCO_2e/kWh$ (2025), kerosene ~0.247, coal ~0.345. Electric routes use the live GB grid carbon intensity from the NESO Carbon Intensity API [21], as a seven-day mean. Bioenergy is counted at zero combustion emissions under the biogenic convention, excluding supply-chain emissions — a stated simplification. Because combustion factors are fixed by fuel chemistry while the electric factor tracks the decarbonising grid, the heat-pump rows fall over time and the combustion rows do not; the panel is designed to make this divergence legible.

7. Indigenous-supply share

The headline indigenous share is computed on a delivered-service basis: each unit of delivered heat or cooling inherits the UK-origin fraction of its energy input — gas ~42% (UK-origin share of the gross supply pool — UKCS production plus grid biomethane over production, imports and stock draw — DUKES 2026 [22]), grid electricity ~75% (†, from generation-mix and interconnector data), harvested ambient and ground heat 100%. The last follows the Eurostat and DUKES convention of treating captured ambient heat as indigenous renewable supply [22,23]. Cooling's delivered multiple inherits the origin share of its input electricity; thermodynamic leverage is not treated as an energy origin. The purchased-energy-basis share is retained in the published data for continuity; it changes little under a geothermal shift because ground heat is never purchased, which is the substantive point the services basis is designed to expose.

8. Other panels

8.1 Non-gas fuels

Oil, bioenergy, solid fuel, heat networks and electric resistive heating lack daily operational feeds and are held at ECUK annual levels [4], modulated where appropriate by degree days. They move slowly and contribute little week-to-week variance.

8.2 Geothermal

The present geothermal contribution is anchored on the EGEC 2025 UK Country Update [24]: approximately 1.43 TWh/yr from an installed base of order 55,000 ground-source heat-pump units, plus of order 0.07 TWh/yr from deep, mine-water and open-loop district schemes (Gateshead, Eden, and others). The stated range is 1.4–2.0 TWh/yr (†); no national register or metering of ground-source output exists, and the Environment Agency's operational-unit estimate is lower than the installed-unit figure, so this is the most weakly constrained quantity on the site. Forward bars for 2027 and 2031 are trend and scenario projections (†) informed by the EGEC Geothermal Market Report 2025 [25] (4,070 UK unit sales in 2025; four new large closed-loop systems; ~11 district systems in development) and the Climate Change Committee Seventh Carbon Budget pathway [26]; the 2050 bar is the Project InnerSpace/REA/ARUP ambition [27]. The European benchmark (2.55 million units delivering 88 TWh in 2025; Swedish sales of 26,785 units against UK sales of 4,070) is drawn directly from [25].

8.3 The empty bar: installed hardware vs the what-if

The 20% what-if is also priced in hardware. Serving one-fifth of delivered buildings heat at 2,000 equivalent full-load hours (†) requires installed thermal capacity of order 30–40 GWth (the exact figure is computed live from the model's delivered-heat estimate); operating UK ground-source capacity is 861 MWth (55,210 systems, EGC 2025 country update), with deeper geothermal adding of order 10 MWth (†, Gateshead, Eden and peers; the EGC table reports only 1.4). The panel sets these against installed reality in France (2,293 MWth ground source + 724 deep), the Netherlands (2,486 + 367 — the Dutch deep capacity being almost entirely greenhouse heat rather than buildings) and Sweden (8,120 + 47), from the EGC 2025 country update summary (Sanner et al., Tables 3–4, end-2024), constants shared with the Irish sibling site. The per-person framing is the substantive point: Sweden operates roughly 778 installed thermal watts per person against the UK's ~13 (†), so the what-if is unremarkable by European precedent and large only in aggregate. A companion calibration restates each installed fleet as a share of its own country's buildings heat (space plus hot water, input basis; national anchors are order-level estimates † — UK live from this model, France ~350 TWh, Netherlands ~115 TWh, Sweden ~80 TWh, per IEA, ODYSSEE and Heat Roadmap Europe sources): the UK sits at ~0.5%, France ~2%, the Netherlands ~5%, and Sweden ~20% — Sweden's existing fleet already serves approximately the share the UK what-if proposes, which is the panel's answer to the scale objection. For capacity flow, the UK replaces of order 1.6 million boilers a year — roughly 35–40 GW of combustion capacity installed annually † — so the what-if corresponds to about one year of the existing replacement flow. Cooling capacity is excluded throughout, consistent with the cross-site scope caveat of Section 8.4's predecessor exchange.

8.4 Northern Ireland

Northern Ireland operates separate gas and electricity systems and the all-island Single Electricity Market, and is not covered by the GB live feeds. Its heat is estimated annually from the Department for the Economy energy statistics and NISRA Continuous Household Survey [28], modulated by NI degree days. Oil heats just over 60% of NI dwellings and has no meter or daily feed; the resulting data gap is itself reported. Northern Ireland is covered properly by the companion site, Irish Heat Split, which applies the same estimator family to the island of Ireland on its own feeds and conventions; the two sites share constants where the underlying quantity is shared and diverge where the systems do, and a cross-calibration between them runs as a regression test in the build checks. Cooling is deliberately not compared across the two: the island figure counts the cold economy (data centres, refrigeration, process), the GB figure counts ECUK comfort cooling and ventilation.

8.5 Whole-economy context (WHY HEAT?)

Four annual pies place heat beside transport and non-heat electricity across delivered services, spend, imported energy and emissions. Sourced anchors are ECUK 2025 final-energy and transport totals [4] and DESNZ provisional greenhouse-gas statistics by sector [29]; the service-level allocations (heat total ~630 TWh, spend, import and emission splits) are author derivations (†), deliberately rounded. Non-energy emissions are excluded; the view is of energy services, not a national inventory.

9. Historical series (the trend layer)

9.1 Construction

From v5.0 the four headline estimates — purchased energy, indigenous share, national bill and emissions — and their 20%-geothermal what-if counterparts are also published as a weekly historical series in the daily data file, rendered on the site as a sparkline under each headline with a selectable one-week to twelve-month window. The series is built from calendar weeks (Monday–Sunday); a week enters only if the gas feed served all seven of its days, so every published point is live by construction — no modelled or interpolated weeks appear. Each point is computed by the same estimator code path that produces the current-week headline figures: one implementation, applied to a different week, rather than a parallel reconstruction that could drift. Entries are frozen as first published, except the two most recent weeks, which are recomputed each day against upstream feed revisions; the series is capped at 60 weekly entries. The stored schema has evolved by documented one-time restatements (a version field records the current schema): each restatement recomputes every stored week still inside the live gas window through the same estimator path, reusing each week's stored grid intensity and the tariff in force, so no historical inputs are refetched or altered — restatements to date added one-decimal precision to the indigenous share and per-week heat/cooling splits of purchased energy, bill and emissions for both the actual and what-if series. Weeks the feed window has rolled past are unrecomputable and retain their original form, stated where it affects a displayed total.

9.2 Time-consistent pricing and carbon

Two inputs vary materially within the window and are applied at their historical, not current, values. Monetary figures for each week use the Ofgem default-tariff cap unit rates in force during that week, from a maintained table of cap periods, so the bill series reflects price-cap movements rather than repricing the past at today's rates. Emissions for each week use that week's mean GB grid carbon intensity, computed as the mean of the half-hourly NESO actuals for the seven days [21], so the electric-route contribution falls through the window as the grid decarbonises. Estimator constants for which no weekly series exists — non-cap fuel prices (heating oil, solid fuel, bioenergy), the non-domestic gas and electricity rates and sector shares of the bill blend (Section 6.2), the ECUK anchor vintage, and the indigenous input-origin shares — are today's values applied uniformly across the window. This is stated alongside the series; it bounds the interpretation of the bill and indigenous-share histories more than the energy and emissions histories, and it means the earliest weeks of the window are not exact reproductions of what the dashboard published at the time.

9.3 The live/modelled boundary

The National Gas operational feed serves approximately the trailing thirteen months of daily LDZ demand. The feed's actual served window is recorded at every build in the published data (oldest and newest dates served, and the day count), so the boundary of the live series is empirical rather than assumed. The trend layer publishes nothing beyond that window. A modelled back-cast — the same regression applied to archived degree days (ERA5 reaches 1940; NESO carbon-intensity archives reach 2018) and calibrated to the ECUK vintages then current — remains a planned second phase and is not built. What v7 delivered instead is depth rather than length: the thirteen-month hourly retrospective of Section 10, built on the same archives, which establishes the fetching, storage, restatement and gating machinery a back-cast would need. Outstanding for the back-cast are the historical ECUK vintages, a per-vintage calibration path, and the distinct visual grammar; live and modelled points will never be blended undistinguished.

9.4 Presentation

Each sparkline shows the actual series as a solid line and the 20%-geothermal what-if as a lighter line, with the band between them shaded: the vertical gap is the estimated forgone saving in that week, and its integral over the window is the cumulative case for the counterfactual. The change since the window start is stated per headline — in percentage points for the indigenous share, in per cent otherwise — and coloured by whether the movement is welcome (a rising indigenous share; falling energy, bill and emissions), not by its sign. The window selector also re-totals the headline figures and their what-if counterparts: sums of purchased energy, bill and emissions over the complete live weeks of the window (four, twelve or fifty-two), the indigenous share as a purchased-energy-weighted average, and the what-if row restated as the cumulative forgone saving; every splittable figure — purchased energy, bill and emissions, actual and what-if — carries its heat/cooling split in each window's caption, summed from the per-week stored splits. At the default this-week view the headline figures are the live estimates and the sparkline shows the full record as context.

10. The hourly retrospective and the system view

10.1 Purpose and construction

The panels of Sections 1–9 describe the transition in weekly aggregate. Section 10 tests it against the electricity system Britain actually operated, hour by hour, across the trailing twelve months. A separate engine reconstructs an hourly retrospective and stores it alongside the weekly file: temperature (ERA5, twelve population-weighted points, identical to the degree-day layer), wholesale price (Elexon market index), grid carbon intensity (NESO), observed underlying demand, and observed wind and solar generation. Thirteen months are held so that twelve complete calendar months always exist for the seasonal exhibit; every annual statement is computed on the trailing 8,760 hours.

10.2 Hourly shaping of heat demand

Annual useful space heat is distributed to days in proportion to each day's share of heating degree days at the live regression base, then within each day in proportion to heating degree-hours; hot water is a flat hourly rate. Two conventions govern the result. First, days below a threshold of one heating degree day contribute no space heat, with a smooth ramp to full contribution at three (†): raw degree-hour allocation assigns winter-strength heat to cool summer nights, which is behaviourally implausible and inconsistent with the weekly estimator, whose baseline subtraction reports no space heating in those weeks. Second, the weights are normalised after damping, so the trailing year still integrates exactly to the annual anchor — the damping redistributes heat into the heating season rather than discarding it. The resulting shape is modelled, not metered, and is marked as such wherever it is drawn.

10.3 Coefficient of performance and calibration

Each replacement route is evaluated hourly as a Carnot fraction: $COP = \eta \times (T\_flow + 273.15) / \max(8, T\_flow - T\_source)$, with source temperatures of outdoor air less a 3 K approach for air-source, 8 °C for shallow ground and 10 °C for geothermal networks (†). Flow temperature is weather-compensated from 30 °C at +15 °C outdoor to 50 °C at −5 °C for space heating, and fixed at 52 °C for hot water year-round (†); each hour's electricity is the load-weighted sum of the two legs. Source temperatures enter every hourly COP directly through the lift term, not as a label: air-source follows the outdoor series hour by hour, ground and network sources are held constant at their stated values. Ground at 8 °C is a UK shallow-borehole annual mean; 10 °C for networks reflects an ambient loop rather than a deep resource, and is conservative — a deeper or waste-heat-fed network would run warmer. That conservatism costs little either way, because the seasonal performance anchor pins the annual: raising the network source from 10 to 15 °C lowers the fitted η to compensate and moves the worst-hour addition from 6.8 to 7.1 GW — that is, a warmer assumed source makes the published winter figure very slightly WORSE, not better, so the choice cannot flatter the case. The flow regime is deliberately the same for all three routes: the what-if is a replacement onto the existing emitter stock, so weather compensation applies whatever the source. A flat performance factor for networks was considered and rejected on that basis, and because it would have improved the figure most favourable to the thesis by roughly a quarter. Air-source additionally carries a defrost band peaking near 2 °C, shape taken from the open monitored fleet at heatpumpmonitor.org and level re-anchored by calibration (†). The scale factor η is then solved per route by bisection so that the trailing year integrates exactly to the published seasonal performance factors of 2.80 (air-source), 3.24 (ground source) and 5.0 (networks, a site convention †). Because hot water is a harder duty than weather-compensated space heating, the split raises η and lowers the winter route loads relative to a single-flow model; it also reverses the summer ordering, air-source outperforming ground source above roughly 11 °C — the temperature at which outdoor air, less its approach, exceeds the ground temperature.

10.4 Stress, netting and the binding hour

Two hourly statistics are published. The first is the coldest hour: the year's highest modelled heat demand, with each route's replacement electricity expressed against that hour's observed demand. It is a gross figure. The second is the binding hour: for each route, the hour maximising observed demand plus the route's net addition less observed wind and solar, searched across November to March. Net here subtracts the fifth of existing resistive space heating the what-if displaces, shaped like space heat and route-independent; that displaced load is itself space-shaped, and night-storage tariff profiles would reduce the credit at morning and evening peaks (†), so the gross figures bound the zero-credit case. The two statistics need not fall in the same hour, or even the same week, and their ratios differ: the coldest-hour figures are additions alone and separate by the ratio of coefficients of performance, while the binding-hour figures are total system requirements dominated by a common standing demand.

10.5 The capacity line

Requirements are set against a capacity line constructed as a fixed de-rated block of dispatchable and interconnector capacity plus the wind and solar actually generated in each hour. The block is taken as 62 GW (†) from the NESO Winter Outlook 2025/26: a base-case de-rated margin of 6.1 GW at 10.0% of average cold spell peak implies peak demand of 61.0 GW and de-rated supply of 67.1 GW, from which the Outlook's own wind and solar de-rating of approximately 5.1 GW is removed to avoid double counting. Taking renewables as observed rather than de-rated is deliberate: it removes the most contested assumption in margin arithmetic, and the line then sags exactly when the wind did not blow. THE BLOCK IS A CONSTANT AND MUST BE REVISED EACH AUTUMN against the incoming Winter Outlook; it is winter-assessed, so all capacity statements are confined to November through March.

10.6 What the capacity comparison is not

It is a static overlay, not a dispatch model. No redispatch, interconnector response, demand-side response or price response is represented; distribution-level constraints are unmodelled; and a capacity expressed in gigawatts cannot represent storage duration — batteries can serve a peak hour but not a fifth consecutive cold day, which is what the exhibit week shows. Exceedance is therefore reported as hours in which a route's requirement exceeds available headroom under today's fleet, never as any statement about supply interruption. One winter is one sample: in 2025-26 the cold snap was windy and the still spell (18–21 March, wind falling to 1.6 GW) was mild, so the coincidence of the two remains a risk the reader may construct but this record does not demonstrate.

10.7 The coincidence premium

Each route's annual replacement electricity is priced twice: at the wholesale index of the hour in which it is drawn, and at the year's flat mean price. The difference is the coincidence premium — the cost of when heating draws power rather than how much. It is small and positive for all three routes, largest for air-source, and it is a wholesale-basis figure that sits beside the daily spark gap rather than in the capped retail bill.

10.8 Verification

Four gates run before any hourly output is published. G1 confirms each route's calibrated performance factor reproduces its anchor. G2 reconciles the hourly network-route electricity against the weekly estimator's equivalent figure; it is near-tautological by construction and therefore proves wiring and heat conservation rather than independent physics. G3 confirms cooling electricity is conserved against its annual anchor. A feed gate refuses any series missing more than five per cent of its hours. Failure of any gate leaves the previous hourly output in place and the weekly panels untouched. Independent corroboration comes from elsewhere: the located worst hour falls inside the peak week that the weekly gas estimator identifies separately, and on the same day as the year's record observed demand hour.

11. Automation, failure handling and reproducibility

A scheduled GitHub Actions workflow executes daily at 03:43 UTC: it retrieves each feed, fits the regressions, computes the balance, and commits a single JSON data file rendered by a static page. No authentication is required for any source; all are open data. The repository is public, so every computation behind every published number is inspectable. The data file carries the weekly trend history of Section 9, so the daily commit log doubles as its backup and audit trail. Each feed is wrapped so that a failure retains the last valid value and flags the affected panel; sources that publish on a lag are shown as provisional when their latest value is stale. Publication identifiers are resolved from source catalogues at runtime, and data units are inferred from magnitude, so that upstream renaming or unit changes surface as diagnostics rather than silent errors. Build failures raise an explicit notification, so that the absence of an alert is itself evidence that the pipeline is current.

12. Principal limitations

The following are the material limitations, stated plainly:

- All live figures are model estimates from temperature regression on operational data calibrated to annual statistics, not metered national heat.
- The gas-to-heat decomposition assumes a stable demand–temperature relation; it degrades in atypical weather and under price shocks, and cannot cleanly separate cooking and non-heating gas.
- The observed and ECUK-anchored cooling estimates differ by an order of magnitude; a year-round regression diagnostic now decomposes the gap, but its weather-flat remainder cannot separate ventilation from base-load cooling (Section 4.5) (†).
- The comfort-deficit stock fractions and thermal-response coefficients carry a several-fold range (Section 5.1).
- Geothermal output is not nationally metered; the present-day bar is the most weakly constrained figure on the site (Section 8.2).
- The historical series covers only the window the live gas feed serves (~13 months); estimator constants without weekly series are applied uniformly across it (Section 9.2).
- Annual anchors (ECUK, DUKES) lag by roughly nine to fifteen months; the live layer is provisional until each annual release.
- Hourly heat shape is modelled from weather, not metered; the summer damping threshold and the flow-temperature conventions are estimates (†, Section 10.2–10.3).
- The capacity line is a static overlay on one winter of observed weather, with a de-rated block that must be revised each autumn; it cannot represent storage duration or network constraints (Section 10.5–10.6).
- Displaced resistive heating is space-shaped; night-storage tariff profiles would reduce the credit at peaks, so gross and net figures bound the truth between them (Section 10.4).

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Every author estimate on the dashboard carries a dagger (†) and an invitation. Corrections and challenges to any figure, source or method in this document are welcomed at contact@causewaygt.com.

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